

Your Cancer Summary Report

Michelle Thompson

Stage IIA Invasive Ductal Carcinoma, ER+/PR+/HER2-

Report Date: May 07, 2026

⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - Invasive Ductal Carcinoma (IDC), Luminal A

Stage: Stage IIA | **Grade:** Grade 2

You have been diagnosed with a Stage IIA breast cancer that started in the milk ducts. This type responds well to hormone-blocking treatment. Your HER2 protein level is low — confirmed across every region of your tumour — which means HER2-targeted drugs are not needed.

✓ Positive Factors

- ER+/PR+ (ICD Z15.01 BRCA2 susceptibility noted) — strong candidate for endocrine therapy; CA 15-3 18 U/mL and CEA 2.1 ng/mL both within normal limits on current tamoxifen
- PIK3CA H1047R at VAF 0.42 — FDA-approved alpelisib targeted therapy available; strongest multi-omics signal across all 3 layers (Stouffer $q=1.57e-10$)
- HRD score 60 (≥ 42 threshold, HRD-positive) — PARP inhibitor therapy eligible
- ERBB2 absent from all 900 spatial spots across 5 regions — definitive HER2-negative confirmation
- Luminal A spatial signature: PGR I=0.739, ESR1 I=0.713, GATA3 I=0.602, FOXA1 I=0.567 — all significantly clustered

⚠ Challenges to Consider

- GATA3 frameshift (VAF 0.31) and CDH1 splice donor (VAF 0.28) —

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- Subclonal TP53 R248W at VAF 0.15 — requires longitudinal monitoring for clonal expansion
- HRD score derived from simplified POC algorithm — clinical-grade Myriad myChoice assay required to confirm PARP eligibility
- Germline BRCA2 not detected in somatic VCF; dedicated germline blood panel required given BRCA2 susceptibility ICD code

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

PIK3CA

Pathogenic

H1047R missense (chr3:179234297 A>G, QUAL 1245.6, depth 238x)

A change in the PIK3CA gene keeps a key growth signal permanently switched on in your cancer cells. This is the most common actionable mutation in Luminal A breast cancer, and it means you are eligible for a targeted drug called alpelisib that specifically blocks this signal.

Treatment Option: Alpelisib + fulvestrant (SOLAR-1 trial): FDA-approved for HR+/HER2- with PIK3CA mutation. Strongest Stouffer signal: $q=1.57e-10$.

GATA3

Likely Pathogenic

Frameshift (chr10:8095656 GCTT>G, QUAL 934.2, VAF 0.31)

The GATA3 gene helps define normal breast cell identity. This change is a hallmark of the Luminal A subtype and confirms your cancer responds best to hormone-blocking therapy.

Treatment Option: Confirms Luminal A — supports endocrine therapy continuation

CDH1

Likely Pathogenic

Splice donor variant (chr16:68771967 G>A, QUAL 678.4, VAF 0.28)

CDH1 makes E-cadherin, a protein that helps cells hold together. A splice site change alters how the protein is produced. This finding does not change your current treatment but is being actively monitored.

Treatment Option: No targeted therapy currently available — clinical monitoring recommended

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MAP3K1

VUS

Q761X stop-gained (chr5:56798505 C>T, QUAL 856.8, VAF 0.35)

A stop signal cuts the MAP3K1 protein short. This is a common finding in Luminal A breast cancers. Its exact clinical impact is not yet fully understood and will be re-evaluated as knowledge grows.

Treatment Option: VUS — no targeted therapy; re-evaluate against updated databases annually

TP53

Likely Pathogenic

R248W missense (chr17:7675088 G>A, QUAL 312.4, VAF 0.15)

A change in the TP53 tumour-suppressor gene was found in about 15% of tumour cells — a small subclonal population. Because it is present in only a fraction of cells, the focus is on careful monitoring rather than a change in your treatment plan right now.

Treatment Option: Subclonal — monitor for clonal expansion at future sequencing timepoints; no specific targeted therapy at this VAF

ERBB2 (HER2)

Wild-type / Negative

Not detected — absent from all 900 spatial spots across 5 tissue regions

HER2 was completely absent from every spot analysed across your entire tumour map. This is the strongest possible confirmation that your cancer is HER2-negative, meaning HER2-targeted drugs such as trastuzumab are not indicated.

Treatment Option: HER2-negative definitively confirmed — trastuzumab/pertuzumab not indicated

HRD (Homologous Recombination Deficiency)

Pathogenic

Score 60 — HRD-positive (LST=60, LOH=0, TAI=0; threshold ≥ 42)

Your tumour has a high level of genomic scarring — a score of 60 on the HRD scale — indicating that cancer cells have difficulty repairing DNA damage. PARP inhibitor drugs are designed to exploit exactly this weakness, making your cancer more sensitive to them.

Treatment Option: HRD-positive: olaparib / niraparib eligible. Confirm with clinical-grade Myriad myChoice assay before initiating.

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

900 spots mapped across 5 tissue regions: tumor_core (305), adipose (287), stroma (186), immune_infiltrate (118), normal_epithelium (4). Moran's I spatial autocorrelation confirmed strong clustering of all 4 Luminal A driver genes. ERBB2 undetected in all 900 spots — definitive HER2-negative confirmation. Immune microenvironment is cold (118 immune_infiltrate spots; limited immune activation expected).

Immune Status: • Low immune activity - The tumor may be evading your immune system

A detailed spatial map of your tumour shows it is strongly driven by oestrogen signals throughout all tissue regions. The HER2 protein was completely undetectable in every analysed spot — the firmest possible confirmation you do not need HER2-targeted drugs. The immune response in your tumour is currently low, which is typical for this type of breast cancer and means immunotherapy is not recommended at this time.

Key Patterns Observed

- PGR: Moran's $I=0.739$ (strongest; $z=459$) — progesterone receptor dominance throughout tumor_core
- ESR1: Moran's $I=0.713$ ($z=443$) — oestrogen receptor significantly clustered, confirms ER+ spatial identity
- GATA3: Moran's $I=0.602$ ($z=374$) — Luminal A transcription factor co-localised with ESR1/PGR
- FOXA1: Moran's $I=0.567$ ($z=352$) — ER pioneer factor co-expressed; Luminal A signature complete
- MKI67: Moran's $I=0.274$ ($z=171$) — proliferative hotspots spatially localised in tumor_core

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- ERBB2: not found in expression data — absent from all 900 spots across all 5 regions
- Outputs: morans_i_plot_20260507_184140.png ·
region_composition_20260507_184132.png

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen 20 mg daily (continue)

Hormonal/Endocrine

FDA Approved

Expected Response: 70-80% in ER+/PR+ patients

Tamoxifen blocks oestrogen from fuelling your cancer cells. You are already taking it and responding well — your tumour markers CA 15-3 (18 U/mL) and CEA (2.1 ng/mL) are both within the normal range, and your haemoglobin and WBC are stable.

Why this may help you: Standard of care for premenopausal ER+/PR+ Stage IIA IDC. Spatial Moran's I confirms strong ESR1 ($I=0.713$) and PGR ($I=0.739$) clustering throughout the tumour, supporting continued hormone sensitivity.

Common side effects: Hot flashes, Mood changes, Blood clot risk, Bone density loss over time

Alpelisib + Fulvestrant

Targeted Therapy

FDA Approved

Expected Response: 35.7% PFS benefit vs. placebo (SOLAR-1)

Alpelisib blocks the overactive PI3K growth signal caused by your PIK3CA H1047R mutation. Fulvestrant blocks oestrogen receptors. Together they cut off two of your tumour's main growth pathways simultaneously.

Why this may help you: PIK3CA H1047R at VAF 0.42 — the highest-frequency, highest-confidence actionable variant. Confirmed pathogenic at chr3:179234297. Stouffer meta $q=1.57e-10$ across all three omics layers: RNA effect size 2.6, protein 2.7, phospho 3.2.

Common side effects: High blood sugar (hyperglycaemia), Nausea, Rash, Diarrhoea, Fatigue

Palbociclib or Ribociclib (CDK4/6 inhibitor)

Targeted Therapy

FDA Approved

Expected Response: 55-60% combined with endocrine therapy

CDK4/6 inhibitors slow the machinery that drives cell division. MKI67 spatial clustering in your tumour confirms active proliferation. Adding this class of drug to endocrine therapy significantly delays cancer regrowth.

Why this may help you: MKI67 Moran's $I=0.274$ (weakly patterned, $z=171$) confirms localised proliferative hotspots. CCND1 multi-omics signal Stouffer $q=1.10e-5$. PALOMA-2 and MONALEESA trials support CDK4/6i in HR+/HER2- disease.

Common side effects: Low white blood cell count, Fatigue, Nausea, Mouth sores, Infection risk

Olaparib or Niraparib (PARP inhibitor)

Targeted Therapy

FDA Approved

Expected Response: ~60% in HRD-positive patients (OlympiAD/BRAVO)

PARP inhibitors block a DNA repair pathway that HRD-positive tumour cells depend on as a backup system. With your HRD score of 60 — well above the threshold of 42 — your cancer cells are especially vulnerable to this type of drug.

Why this may help you: HRD score 60 (LST-driven genomic scars) exceeds the clinical threshold of 42. Patient ICD code Z15.01 (BRCA2 susceptibility) on record. Clinical-grade Myriad myChoice assay and dedicated germline BRCA2 panel required to confirm eligibility before initiating therapy.

Common side effects: Fatigue, Nausea, Anaemia, Low platelets, Shortness of breath

Clinical Trials

You may be eligible for these research studies offering new treatments:

SERENA-6: Camizestrant + CDK4/6i in HR+ Breast Cancer with PIK3CA Mutation

Phase: 3 | **Status:** Recruiting

Trial ID: NCT03709082

Your PIK3CA H1047R mutation at VAF 0.42 is exactly the population targeted by this Phase 3 trial of a next-generation oestrogen receptor degrader combined with CDK4/6 inhibition.

Nearest Location: Multiple US academic cancer centres

Contact: ClinicalTrials.gov NCT03709082

OlympiA Extension: Adjuvant Olaparib in HRD-Positive Early Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT04592783

Your HRD score of 60 meets the genomic eligibility threshold for adjuvant PARP inhibitor therapy in early-stage breast cancer — ask your oncologist whether you qualify.

Nearest Location: Nationwide US sites

Contact: Ask your oncologist about OlympiA adjuvant olaparib eligibility criteria

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months	CA 15-3 + CEA tumour markers Track response to tamoxifen and detect early recurrence — currently WNL (CA 15-3 18, CEA 2.1)
Every 6 months	Mammogram + bilateral breast MRI Imaging surveillance — critical given HRD-positive profile and Z15.01 BRCA2 susceptibility code
Every 3 months if alpelisib started	Fasting glucose and HbA1c Monitor for hyperglycaemia — primary adverse effect of alpelisib; baseline Hgb 12.5 g/dL noted
Annually	Bone density scan (DEXA) Tamoxifen-associated bone density loss monitoring
Once, as soon as possible	Clinical-grade HRD assay (Myriad myChoice) Validate HRD score 60 with approved clinical assay to confirm PARP inhibitor eligibility
Once within 3 months	Germline BRCA1/2 blood panel Confirm germline BRCA2 status per ICD Z15.01 code — guides both PARPi eligibility and family counselling
At next disease progression or 12 months	TP53 subclonal monitoring (repeat sequencing) Monitor TP53 R248W VAF 0.15 for clonal expansion

⚠ When to Call Your Doctor

Contact your care team right away if you experience:

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- Persistent bone pain or fracture without injury
- Shortness of breath, persistent cough, or chest pain
- Unexplained weight loss greater than 5 kg
- Excessive thirst, frequent urination, or blurred vision (hyperglycaemia if on alpelisib)
- Unusual bruising, prolonged bleeding, or severe fatigue (if on PARP inhibitor or CDK4/6i)

Questions? Your oncology nurse navigator or Dr. Sarah Chen's office — contact immediately if any warning signs appear

For Your Family

Your medical record includes a BRCA2 genetic susceptibility code (ICD Z15.01), and your tumour's HRD score of 60 is consistent with an underlying germline DNA repair defect. A dedicated germline blood panel is required to confirm. If a BRCA2 pathogenic variant is found, each first-degree relative (parents, siblings, children) has a 50% chance of carrying it, with substantially elevated lifetime risks for breast cancer (up to 69%) and ovarian cancer (up to 17%).

Recommended Steps for Family Members

- Order dedicated germline BRCA1/2 blood panel — distinct from somatic tumour sequencing
- First-degree relatives (daughters, sisters, brothers) should discuss BRCA genetic testing with their own physicians
- If germline BRCA2 confirmed: discuss prophylactic bilateral salpingo-oophorectomy with gynaecological oncologist
- Sons at elevated risk for male breast cancer and prostate cancer — testing recommended if germline confirmed

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:



FORCE — Facing Our Risk of Cancer Empowered

National organisation for individuals with hereditary breast and ovarian cancer — highly relevant given BRCA2 susceptibility (Z15.01) and HRD-positive profile

📞 1-866-288-7475 | 🌐 <https://www.facingourrisk.org>



Breastcancer.org

Plain-language guides, community forums, and expert Q&A for all breast cancer diagnoses and treatment decisions

📞 1-855-4BREAST | 🌐 <https://www.breastcancer.org>



NSGC Genetic Counsellor Finder

Locate a certified genetic counsellor to guide germline BRCA2 confirmation and family risk discussions

| 🌐 <https://www.nsgc.org/find-a-gc>

⚠ Important Notice

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Data Sources: mockepic — PAT002-BC-2026 demographics/labs/meds parsed: Michelle Thompson, 42F, CEA 2.1, CA 15-3 18, tamoxifen 20mg, ICD Z15.01 BRCA2 susceptibility, fgbio — PAT002_tumor_R1.fastq.gz not found at absolute path (FASTQ not yet generated), genomic-results — PAT002_somatic.vcf parsed: 5 variants (PIK3CA H1047R VAF 0.42, GATA3 fs 0.31, CDH1 splice 0.28, MAP3K1 Q761X 0.35, TP53 R248W 0.15), genomic-results — PAT002_cnv.cns parsed: 28 neutral segments; HRD score 60 (positive, LST=60), spatialtools — PAT002_regions.csv: 900 spots, 5 regions; Moran's I: PGR=0.739, ESR1=0.713, GATA3=0.602, FOXA1=0.567, MKI67=0.274; ERBB2 not found, multiomics — RNA(1000) + protein(400) + phospho(250), 12 samples integrated; Stouffer meta all 5 BC genes significant (ESR1 $q=8.3e-9$, PGR $q=5.5e-7$, PIK3CA $q=1.6e-10$, CCND1 $q=1.1e-5$, MKI67 $q=2.4e-9$)

Report Version: 1.0

Generated: 2026-05-07 18:43:25

